

Nkateko Tibane

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Professional Summary

Final-year BSc Information Technology student with a strong focus on penetration testing, vulnerability research, exploit development, and systems programming. Experienced in researching software vulnerabilities, developing proof-of-concept exploits, building offensive security tooling in C/C++, and conducting low-level analysis of Linux and Windows systems. Passionate about security research, continuous learning, and applying technical knowledge to identify and remediate real-world security weaknesses.

Highlighted Personal Projects and Experience

Vulnerability Research & Technical Root-Cause Analysis

Independent Security Analyst

- Performed root cause analysis of publicly disclosed vulnerabilities involving memory corruption, integer overflows, and heap-based vulnerabilities.
- Developed proof-of-concept analyses explaining vulnerability impact, exploitation primitives, and recommended remediation strategies.
- Published technical write-ups translating complex vulnerability research into clear, structured reports for engineers and security practitioners.

Selected Research

- Exploring a Integer Overflow that leads to heap OOB R/W in a GUFF parse: RCA CVE-2025-53630
- Inside CVE-2024-22857 : RCA of zlog - Memory Corruption Bug

pwntrace – Dynamic Instrumentation & Process Monitoring Utility | November 2025 - February 2026

Systems Programmer

- Developed a Linux process instrumentation tool using the `ptrace` API to monitor application execution, inspect memory, and intercept system calls in real time.
- Built features to support reverse engineering, vulnerability research, and exploit development by automating runtime analysis and process inspection.
- Designed a modular command-line utility in C with a focus on Linux internals, low-level debugging, and extensibility for security research workflows.=

TibaneC2 – Command & Control Framework | May 2025 - January 2026

Lead Developer & Architect

- Wrote a multithreaded and modular command and control framework (in C and C++) consisting of a command-line based server and both windows and linux implants that are designed to emulate threat actor infrastructure during security testing and lab environments and data transfer models.
- Implemented encrypted communications and multi-client session management while focusing on secure network programming and offensive security concepts.

Active Capture The Flag (CTF) Player

- Regular participant in offensive security competitions involving web and binary exploitation, reverse engineering and privilege escalation with dead pirate society.
- Developed exploit scripts and documented technical write-ups describing vulnerability discovery, exploitation techniques, and defensive mitigations.
- Collaborated with teammates to solve complex security challenges under time constraints.

Selected Work:

- **Nexhunt CTF Write-up:** <https://t1b4n3.github.io/post/nexhunt-ctf/>
 - **Live Binary Exploitation Walkthrough (YouTube):** https://www.youtube.com/live/jaIR_UykI4w
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Education

Bachelor of Science in Information Technology (BSc IT)

Richfield Graduate Institute of Technology | *Expected Graduation: 2026*

- **Academic Profile:** Currently completing final year of study (2026 graduation target); Strong foundational academic standing.
 - **Relevant Coursework:** Operating Systems, Computer Architecture, Data Structures & Algorithms, Networking, Cloud Computing, IoT Security.
 - **Key Competencies Gained:** Solid systems architecture understanding, low-level testing and debugging, and secure software development lifecycle fundamentals.
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Technical Skills

- **Programming:** C, C++, Python, Java, Bash, SQL
- **Operating Systems:** Linux, Windows
- **Offensive Security:** Penetration Testing, Vulnerability Research, Binary Exploitation, Reverse Engineering
- **Networking:** TCP/IP, HTTP/HTTPS, DNS, Active Directory